

Problem-Solution Fit

Date	1 July 2026
Team ID	xxxxxx
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] - Government disaster management authorities - People living in flood-prone areas - Weather monitoring organizations - Emergency response teams	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited internet access in rural areas - Lack of technical awareness - Insufficient real-time environmental data - High cost of advanced monitoring systems	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Pros: - Basic flood alerts available - Government monitoring exists Cons: - Less accurate prediction - Slow response time - Limited real-time analysis
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Sudden floods causing damage to life and property - Lack of accurate early flood prediction - Delay in warning systems reaching people - Difficulty in monitoring weather changes continuously	9. PROBLEM ROOT / CAUSE [RC] - Unpredictable weather conditions - Inadequate flood monitoring systems - Delay in collecting and processing environmental data - Lack of intelligent prediction models	7. BEHAVIOR + ITS INTENSITY [BE] - People ignore early weather warnings sometimes - Communities react only during emergency situations - High urgency during heavy rainfall seasons

3. TRIGGERS TO ACT [TR] • Heavy rainfall alerts detected • Rising river water levels • Climate change causing unpredictable floods • Need for faster disaster preparedness	10. YOUR SOLUTION [SL] • Machine Learning based flood prediction system • Analyzes rainfall, humidity, river level, and historical flood data • Predicts flood risk in advance • Sends early warnings for disaster prevention	8. CHANNELS OF BEHAVIOR [CH] ONLINE Mobile apps, weather websites, SMS alerts, dashboards
4. EMOTIONS BEFORE / AFTER [EM] Before: Fear, uncertainty, panic, financial loss After: Safety, confidence, preparedness, quick response		OFFLINE Government announcements, local disaster management teams, radio alerts